

4. What are the different errors that can occur in DMA?

- Dangling Pointer
- NULL Pointer
- Garbage
- Double free error

6. C program to create memory for int, char and float variable at run time.

```
#include <stdio.h>

#include <stdlib.h>

int main()
{
    int *ptr_1; // Declare an integer pointer
    char *ptr_2; // // Declare an char pointer
    float *ptr_3; // Declare an float pointer

    // Now allocating memory to each pointer using dynamic memory allocation
    ptr_1 = (int*)malloc(1*sizeof(int));
    ptr_2 = (char*)malloc(1*sizeof(char)*1);
    ptr_3 = (float*)malloc(1*sizeof(float));

    printf("\nEnter the value for integer pointer : ");
    scanf("%d",ptr_1);

    printf("\nEnter the value for char pointer : ");
    scanf(" %c",ptr_2);
```

```
printf("\nEnter the value for float pointer : ");  
  
scanf("%f",ptr_3);  
  
printf("\nThe value stored in integer pointer is : %d",*ptr_1);  
  
printf("\nThe value stored in char pointer is : %c",*ptr_2);  
  
printf("\nThe value stored in float pointer is : %f",*ptr_3);  
  
free(ptr_1);  
  
free(ptr_2);  
  
free(ptr_3);  
  
ptr_1 = NULL;  
  
ptr_2 = NULL;  
  
ptr_3 = NULL;  
  
return 0;  
  
}
```